

Project Abstract: DLA-python Performance Optimization

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Background

DLA-python is a repository for simulating the Diffusion Limited Aggregation process and generating fractal structures. Its core function is simulating random-walking particles to build fractal patterns and measure their properties.

The main problem is slow performance due to many Python loops. This includes: checking collisions one step at a time, rebuilding all sticky sites after each particle, and using nested loops for density calculations. We plan to optimize it from the perspectives of both parallelization and algorithm/data layout optimization. Potential approaches include: using NumPy vectorization to enable parallel array operations, restructuring data structure for better memory access patterns, and applying incremental updates to reduce redundant calculations.

Project Timeline

Week 1: Analysis & Profiling

- Read and understand the codebase
- Use profiling tools to find the slowest parts
- Test baseline performance

Week 2: Parallelization

- Vectorize collision detection and path generation
- Test correctness and measure speedup

Week 3: Algorithm Optimization

- Change sticky site update to incremental method
- Optimize radial density calculation
- Experiment with different data layout options

Week 4: Data Layout Optimization & Documentation

- Restructure data storage for better memory access
- Run full benchmarks and record final speedup
- Write optimization report